

PAVAN KUMAR U

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PROFESSIONAL SUMMARY

Connected Vehicle Engineer with **4+ years in automotive validation**, focused on **HIL/SIL testing, test automation, and requirements-based verification**. Hands-on experience with **Infotainment and Instrument Cluster ECUs**, leveraging **Python, ECU Test tool, GitHub Actions, and CI/CD pipelines** to enhance **test coverage, defect detection, and software reliability**.

EDUCATION

Bachelor of Technology – Electronics and Communication Engineering
Newton's Institute of Engineering, JNTU – Kakinada | CGPA: 7.04

TECHNICAL SKILLS

- CI/CD Platforms & Development Tools: CI Pipelines, Git, GitHub Actions, PyCharm, Python-based Frameworks (ENNA), GCP, Jfrog Artifactory, Swagger, looker studio, Firestore
- Programming & Scripting: **Python**, C
- Operating Systems & Platforms: Embedded Systems, Goldfish/Cuttlefish Emulator
- Automotive Testing Tools: **ECU Test Tool**, Vector, dSPACE ControlDesk
- Test Management & Documentation: IBM DOORS, Code Beamer, IBM RQM, Confluence, AWS
- Communication Protocols: **CAN**, CAN FD, Automotive Ethernet, **UDS** (ISO 14229)
- Test Environments: HIL, SIL, **HIL Test Bench**, AUTOSAR Classic
- Defect Tracking Tools: JIRA, RTC

EMPLOYMENT EXPERIENCE

Verification & Validation Engineer
KPIT Technologies Pvt. Ltd., Bengaluru | Nov 2021 – Present
Leading vehicle software integration, **SIL/HIL validation, CI-integrated test automation, and continuous test execution pipelines for Instrument Cluster and Infotainment (IVI) ECUs**, improving **test coverage, early defect detection, and release readiness**.

PROJECT DETAILS

Client: CARIAD SE Volkswagen Group	July 2025 - Present
Project: CI/CD Automation IVI	
Environment: CI-Integrated SIL / HIL	
<u>Roles and Responsibilities:</u>	
✓ Developed and executed automated test cases using GitHub Actions on the Goldfish Emulator, enabling scalable CI-based validation of IVI features. ✓ Built and maintained CI workflows using ENNA-based Python automation framework, supporting infotainment (IVI) functional and system-level test scenarios. ✓ Executed CI jobs across Dev and Beta environments, ensuring consistent test coverage and early defect detection across release stages. ✓ Integrated automated test execution with RADAR dashboards, publishing results and enabling real-time quality monitoring and reporting. ✓ Automated validation for HCP3 and ICC ECUs, covering ECU communication, system behavior, and infotainment use-case verification. ✓ Performed Smoke, Sanity, Nightly, and Official Build testing to ensure baseline software stability before release promotion	
<u>Milestones:</u>	
✓ Enabled automated execution and result evaluation for IVI test suites using Python, improving regression efficiency. ✓ Ensured continuous quality assurance in CI pipelines, contributing to stable and reliable software releases	

Client: STELLANTIS International Pvt Ltd, Bengaluru

May 2022 - June 2025

Project: R2PX – Vehicle Software Integration (Instrument Cluster)

Environment: HIL

Roles and Responsibilities:

- ✓ Validated and ensured accurate display of alerts and system information on the **Instrument Cluster**.
- ✓ Analyzed Vehicle Function (VF) and requirement documents to create efficient test cases.
- ✓ Designed, developed, and executed manual and automated test cases using IBM Rational Quality Manager (RQM) and **ECU-TEST Tool** for comprehensive test coverage and automation
- ✓ Performed **component-level** testing using **Vector CANoe** to validate ECU functionality and communication protocols.
- ✓ Executed **subsystem-level** testing utilizing **dSPACE ControlDesk** for system validation and calibration.
- ✓ Reported defects using **RTC** and coordinated with development teams for prompt resolutions.
- ✓ Performed Regression, Delta Validation, and Sanity Testing to maintain system stability.

Milestones:

- ✓ Successfully designed and executed 500+ test cases, developed HIL benches, and automated test case migration to boost testing efficiency.
- ✓ Reported 50+ critical defects, improved requirement traceability, and streamlined validation using reusable scripts

Client: KPIT Technologies Ltd, Bengaluru

November 2021 - April 2022

Project: MW Development Project

Environment: AUTOSAR Classic

Roles and Responsibilities:

- ✓ Configured **IPCOM** and **IPCABS** modules based on project requirements and design specifications.
- ✓ Performed code compilation and ensured the integrity of the build process.
- ✓ Conducted unit testing and component testing to verify the functionality of each module.
- ✓ Performed MISRA checks to ensure compliance with coding standards and safety requirements.
- ✓ Executed generation tool testing to validate the outputs from the code generation process.
- ✓ Ensured full traceability of requirements and implementation from design to testing phase.

Milestones:

- ✓ Successfully configured IPCOM and IPCABS modules, meeting all project requirements and improving system functionality.
- ✓ Improved validation efficiency by implementing reusable test scripts and enhancing traceability across the V-cycle

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

PAVAN KUMAR U
Bengaluru